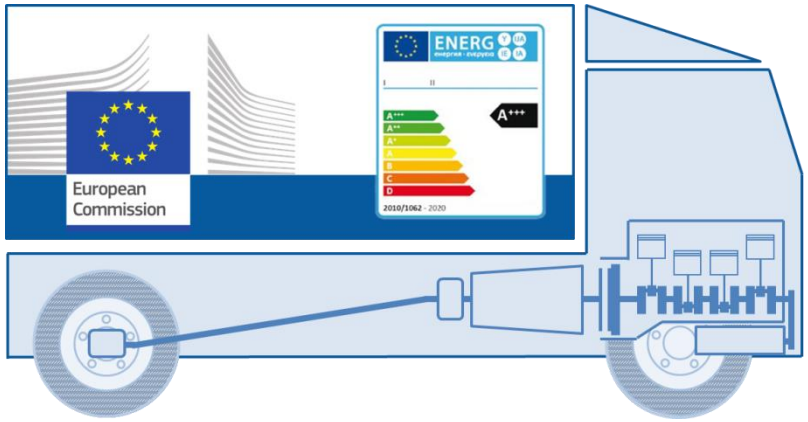
# VECTO Engine pre-processing tool

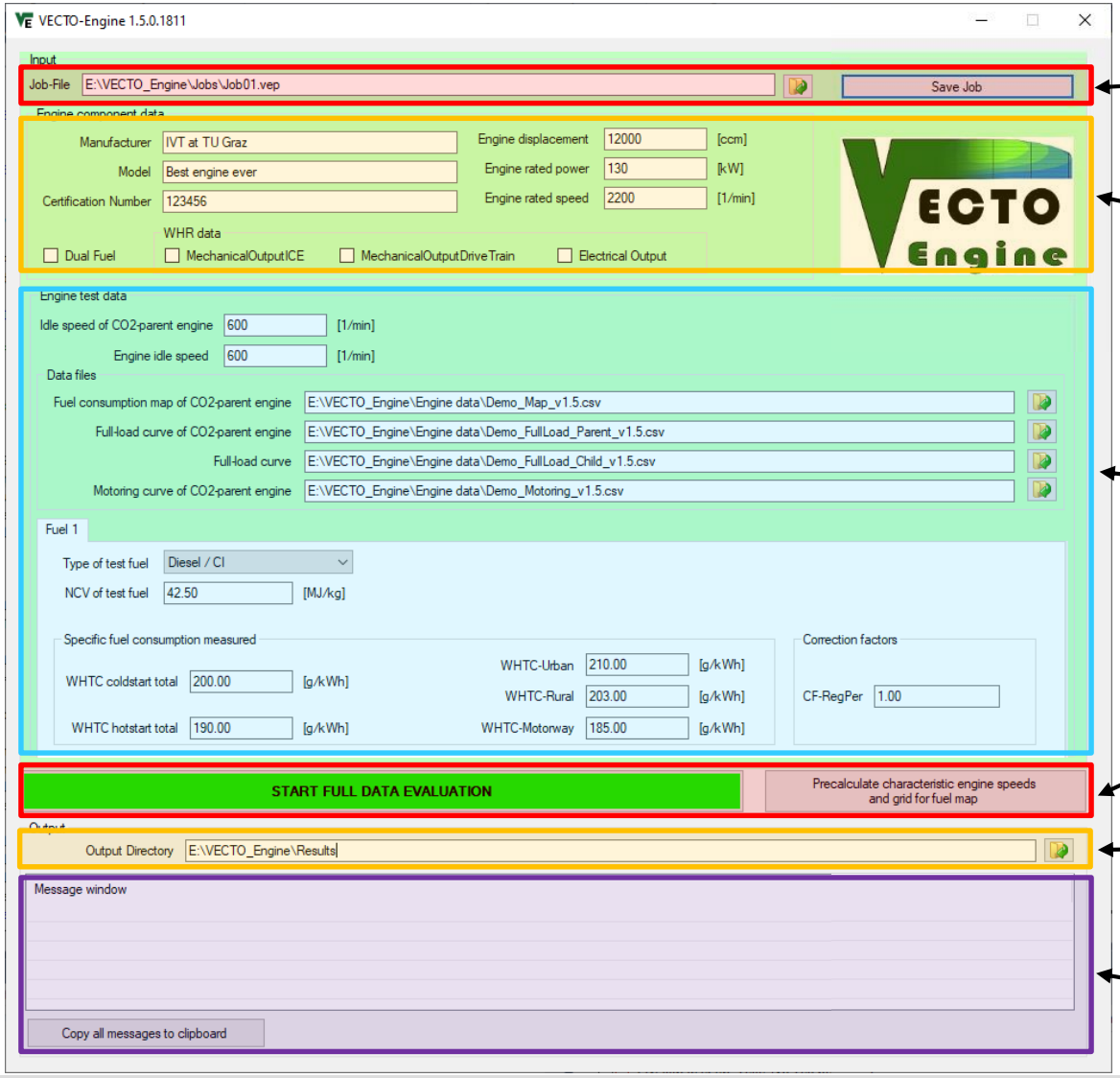
# Short documentation and user manual

## App Version 1.5.*

**01.06.2022**



# VECTO Engine – GUI overview




**Job-file block**

Displays path and filename of job file

Open and save job file

**Component block**

Fields for component information

Checkboxes for technologic features

**Test data block**

Fields for component data

Path and filename of test data files

Test fuel data

Specific fuel consumption figures

Specific WHR work figures (for engines with WHR systems)

**Control button block**

Start data evaluation routines

**Output block**

Displays path of output directory

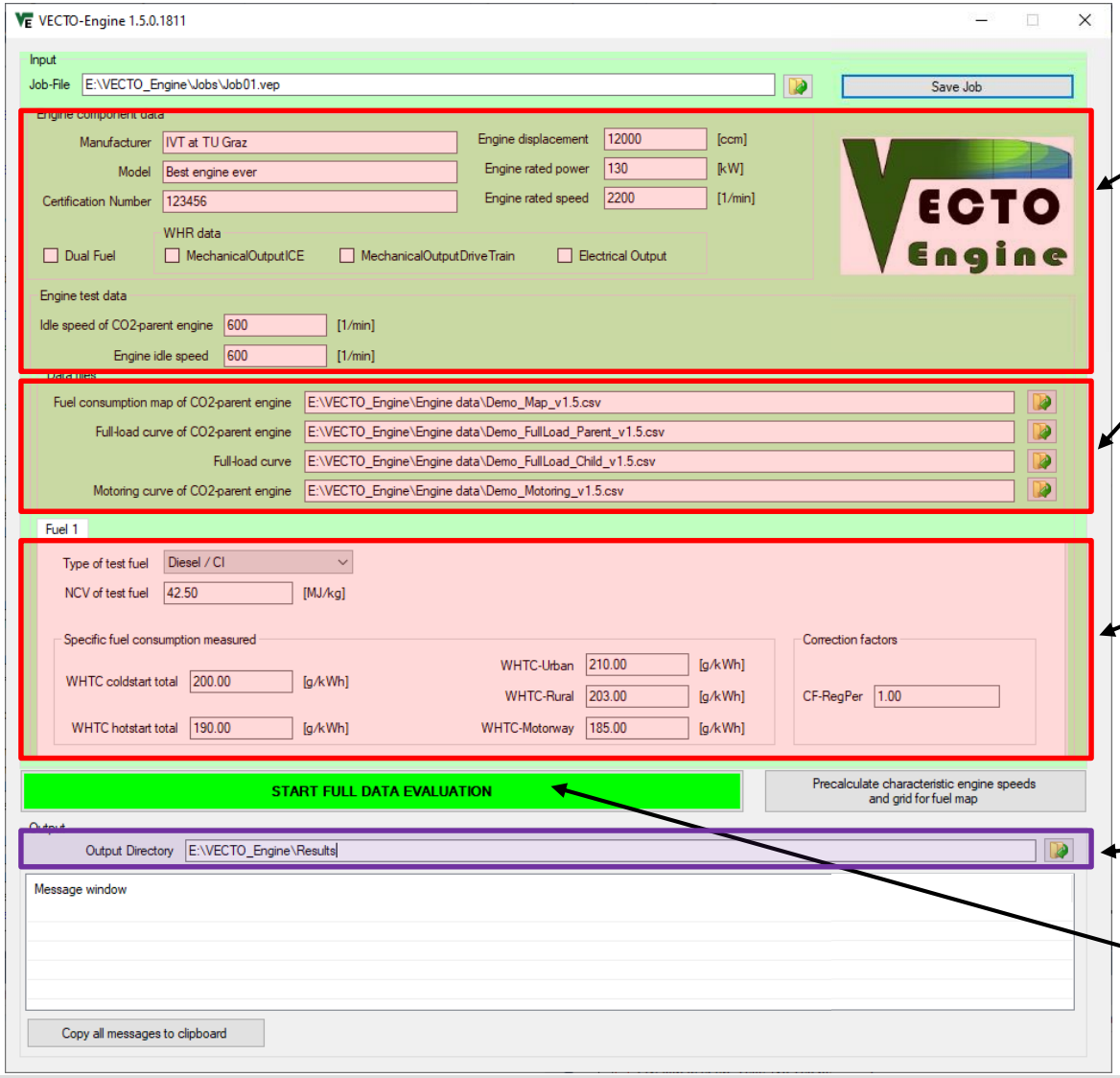
Browse for output folder

**Message window block**

Displays messages during execution

Export messages to clipboard

# VECTO Engine – Operation via GUI




**Step 1**

Input all component data

**Step 2**

Load all required input files via separate open file buttons
(for file format please see demo files)

**Step 3**

Input measured specific FC values and correction factor

→ separate tabs appear depending on setting of technology checkboxes under „engine component data“

*OPTIONAL INPUTS:*

- *Measured specific WHR work (for engines with WHR systems)*
- Measured specific FC values and correction factor for secondary fuel (for dual-fuel engines)

**Step 4**

Define directory where output files are stored

**Step 5**

Press „Start“ button for full evaluation

*(OR press other button to precalculate grid for fuel map from CO2-parent full-load curve)*

# VECTO Engine – Basic features

- Written in VB.net
- Portable application (i.e. no installation required)
- File format conventions *[§6 of Annex V of Regulation (EU) 2017/2400]*
  - CSV (Comma Separated Values) file type
    - List separator is Comma “,”
    - Decimal separator is Dot “.”
- According to Regulation (EU) 2017/2400 the use of VECTO Engine is mandatory
- Output is one single XML file
  - File already hashed
  - Ready for use in VECTO vehicle simulation for declaration

# VECTO Engine – Input files

- VECTO Engine needs 4 separate input files
  (determined according to the current version of the technical annex):
  - Fuel consumption map of CO2-parent engine
  - Full-load curve of CO2-parent engine
  - Full-load curve of engine to be certified
  - Motoring curve of CO2-parent engine

- Input file formats
  - For file formats please refer to the technical annex or supplied demo files!
    ***[§6 of Annex V of Regulation (EU) 2017/2400]***

- Demo data supplied
  - Two different full-load curves for two power ratings of one engine CO2-family supplied
    - Child: full-load of lower power rating
    - Parent: full-load of highest power rating
  - Four different fuel consumption map files plus respective job files (*.vep) supplied
    - Standard single fuel engine
    - Dual-fuel engine w/o WHR system
    - Dual-fuel engine w/ WHR system w/ mechanical output to vehicle's drivetrain
    - Dual-fuel engine w/ WHR system w/ electrical output

# VECTO Engine – Output files

- VECTO Engine produces one XML output file:
  - File name is created based on GUI input data according to the following scheme:

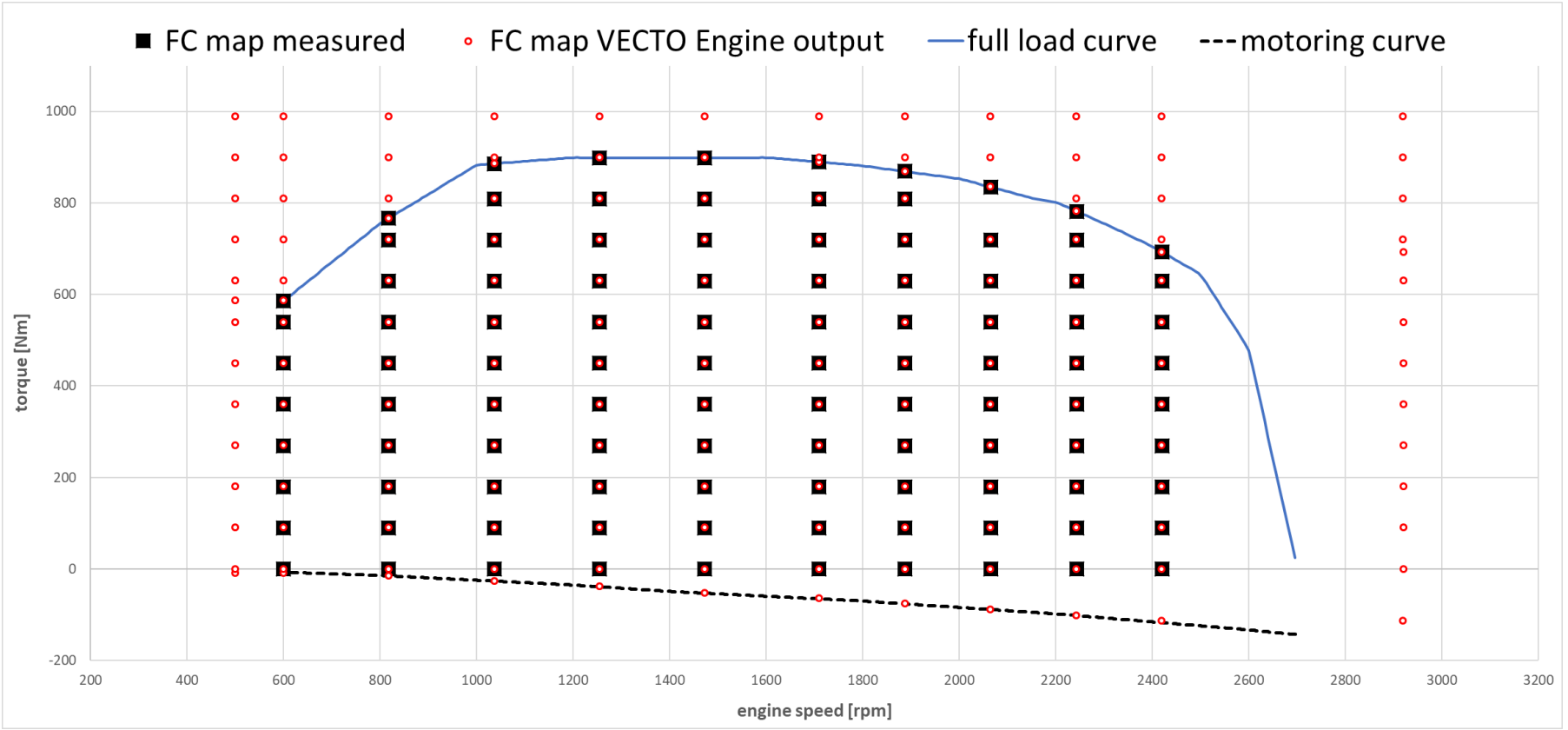
    “Manufacturer” & "_" & “Model” & ".xml“
  - File already hashed
  - Ready for use in VECTO vehicle simulation for declaration

# VECTO Engine – Evaluation algorithms performed (p1/2)

For a detailed description of all operations performed during data evaluation please refer to Appendix 8 to Annex V of COMMISSION REGULATION (EU) 2017/2400

# VECTO Engine – Evaluation algorithms performed (p2/2)

Example of data processing through VECTO Engine for FC map



# VECTO Engine – Further important remarks

- Messages displayed
  - Clear explanations for all relevant errors (ref. to regulation or indication of datapoint)
  - Warnings vs. Errors:
    - A valid XML results file is only created if no errors occur (non-conformity with requirements in regulation)
    - Warnings give a hint that something might be wrong with the input data
      → XML result file is valid despite warnings
      → BUT: warnings should be investigated together with TAA / technical service

- Only one single XML file is created as output
  - File already hashed
  - Ready for use in VECTO vehicle simulation for declaration

# Contact data

- All questions, user support and bug reports need to be handled via the JIRA issue tracking in the CITnet service of the EU-COM: https://citnet.tech.ec.europa.eu/CITnet/jira/projects/VECTO/issues

- Also future updates of the tool will be released only via this platform: https://citnet.tech.ec.europa.eu/CITnet/confluence/display/VECTO/VECTO+Engine+Releases

- If you do not already have an account for this service please contact: JRC-VECTO@ec.europa.eu